



THE ROMAN CATHOLIC DIOCESE OF SUMBAWANGA

**FORM FOUR MOCK EXAMINATION FOR
DIOCESAN SECONDARY SCHOOLS 2026**



031/2A

PHYSICS 2A

Time 2:30 Hours

2026

Instructions

1. This paper consists of **two (2)** questions. Answer **all** the questions.
2. Each questions carries **twenty five (25)** marks
3. All writings should be in **blue** or **black** ink, except for diagrams which must be in pencil.
4. Mathematical tables and non-programmable calculators may be used.
5. Communication devices and any unauthorized materials are not allowed in the examination room
6. Write your **Examination Number** on every page of your answer booklet

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1. You are provided with a metre rule, retort stand and clamp, pendulum bob stop watch and cotton thread. Proceed as follows
 - (a) With a help of thread, tie a pendulum bob to the retort stand and then allow it to touch the floor by adjusting the thread so that the height (H) from the floor to the point of suspension is 100cm. Use two wood pads to tie tight the thread.
 - (b) Adjust the thread so that the bob is at a distance (d) of 10cm from the floor.
 - (c) Allow the pendulum to oscillate for a small and record the time for 20 complete oscillations
 - (d) Repeat the procedure (b) and (c) for d = 20cm, 30cm, 40cm, and 50cm.
 - (e) Record your data for d, t, T and T² in tabular form
 - (f) Plot a graph of T² against d
 - (g) Determine the slope of your experiment
 - (h) Given that $T = 2\pi \sqrt{\frac{l}{g}}$, Calculate the acceleration due to gravity (g).
 - (i) State two possible sources of error in this experiment.
 - (j) On your graph read the d axis and explain its meaning.
 - (k) Suggest the aim of the experiment.

(25 marks)

2. In this experiment you are provided with two plane mirrors, one optical pin, a sheet of plane paper, drawing boards, protractor and mirror holder. Proceed as follows:

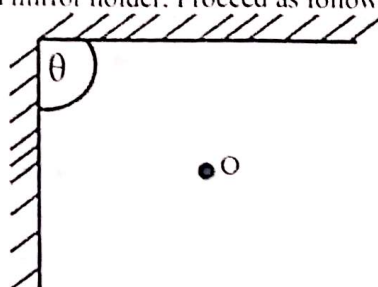


Figure 2

Draw two lines at right angles by using your ruler and protractor on the plane paper and the plane sheet of paper securely fixed on the drawing board. Place one mirror on each line of the sheets of paper. Fix an optical pin on point O and look into both mirrors and count the number of images and record the total number of images from $\theta = 90^\circ$. Repeat the procedure for $\theta = 72^\circ, 60^\circ, 45^\circ$ and 30°

- (a) Tabulate your results as follows:

Number of images, n	Angle between mirrors, θ (deg)	$\frac{1}{\theta}$ (deg ⁻¹)

- (b) Plot the graph of n (vertical axis) against $\frac{1}{\theta}$ (horizontal axis)
- (c) From your graph

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25.70

- (i) Determine the slope of the graph, G to the nearest tenth
 - (ii) Determine n -intercept of the graph, P
 - (iii) Equation relating to n with θ and P is given by $n = \frac{G}{\theta} + P$. Rewrite it with the values of G and P you have calculated.
- (d)
- (i) What happen to the number of images when $\theta = 0^\circ$
 - (ii) Suggest the aim of this experiment
- (25 marks)